SQL Capstone Project

Employee Success Analytics

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NextGen Corp.

POD LUNA

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NextGen Corp.

Business Overview

- NextGen Corp. is a fast-growing technology company specializing in the development of innovative software and hardware solutions. The company is committed to attracting top talent and fostering a high-performance culture that drives continuous growth and innovation.

Business Problem

- Despite its success, NextGen Corp. is currently experiencing key workforce challenges, including increasing employee turnover, inconsistent performance across departments, and salary disparities. Addressing these issues is critical to sustaining employee satisfaction, improving productivity, and maintaining a competitive edge in the industry.

NextGen Corp.

Keys

Total Employees	60
Department_id	**Department Name**
1	Engineering
2	Sales
3	Marketing
4	HR

Employee Retention Analysis

Goal: Understand the employee turnover trends and identify the root causes of high turnover rates

Top 5 highest serving employees?

The findings reveal that a significant proportion of long-tenured employees were hired in 2015, with the Sales department exhibiting the strongest retention. Higher retention among senior and specialized roles suggests that opportunities for advancement and role stability are key drivers of tenure. Conversely, the underrepresentation of other departments among the top performers points to potential retention gaps beyond Sales and Marketing.

```sql
-- Who are the top 5 highest serving employees
select
    e.employee_id,
    e.hire_date,
    current_date - e.hire_date as tenure_days
from employee e
left join turnover t
    on e.employee_id = t.employee_id
where t.employee_id is null
order by tenure_days desc
Limit 5
```

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	employee_id [PK] integer	hire_date date	tenure_days integer
1	8	2015-06-30	3939
2	44	2015-10-27	3820
3	11	2016-06-12	3591
4	26	2016-07-19	3554
5	99	2016-10-13	3468

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What is the turnover rate for each department?

Analysis

- Marketing and Engineering departments experience significantly higher turnover rates, with Marketing department showing critical attrition levels (92.86%). In contrast, Sales and HR departments, which demonstrate stronger retention.

Recommendation

- Immediate intervention is required in high-turnover departments, alongside a review of compensation, management practices, and employee engagement strategies.

```sql
select
d.department_id,
d.department_name,
count(t.turnover_id) as "Employees who left",
count(e.employee_id) as "Total Employees",
 ROUND(
    (CAST(COUNT(t.employee_id) AS numeric)/ NULLIF(COUNT(e.employee_id), 0)) * 100,2) AS "Turnover Rate"
from department d
left join employee e
on d.department_id = e.department_id
left join turnover t
on e.employee_id = t.employee_id
GROUP by d.department_name, d.department_id
order by "Turnover Rate" desc;
```

	department_id [PK] integer	department_name character varying (30)	Employees who left bigint	Total Employees bigint	Turnover Rate numeric
1	3	Marketing	13	14	92.86
2	1	Engineering	4	6	66.67
3	2	Sales	8	29	27.59
4	4	HR	3	11	27.27

Which employees are at risk of leaving based on their performance?

No employee scored below 3.0, indicating consistent baseline performance across the organization. However, the lack of variation suggests possible rating inflation, limiting the ability to identify top and low performers. But going with this rating the employees with id 7, 8, 12, 21, 22, 41 are risk of leaving.

Recommendations

- Revamp Performance Evaluation System
- Introduce a wider rating scale or stricter criteria
- Encourage honest and data-driven assessments
- Motivate employees to exceed expectations

Combine Performance with Other Metrics such as:

- Attendance
- Productivity
- Output metrics

To get a **holistic view of employee performance**

```sql
SELECT
e.employee_id,
e.first_name || ' ' || e.last_name AS "Employee Name",
e.job_title,
p.performance_score
FROM employee e
JOIN performance p
ON e.employee_id = p.employee_id
LEFT JOIN department d
ON e.department_id = d.department_id
where p.performance_score <=3
GROUP BY e.employee_id,e.first_name,e.last_name,e.job_title,p.performance_score;
```

	employee_id integer	Employee Name text	job_title character varying (30)	performance_score numeric (2,1)
1	7	John Brown	Marketing Specialist	3.0
2	8	David Moore	Sales Representative	3.0
3	12	Eve Davis	HR Specialist	3.0
4	21	Grace Smith	Sales Manager	3.0
5	22	Grace Wilson	Sales Representative	3.0
6	41	Charlie Smith	Sales Manager	3.0
7	63	Frank Lee	Sales Representative	3.0
8	98	Frank Wilson	Marketing Specialist	3.0

What are the main reasons employees are leaving the company?

The dominant driver for turnover is **Personal reasons**

Insights

The reasons for turnover can be grouped into three categories:

- **Preventable(** Found another job, Career growth)
- **Partially Preventable(** Personal)
- **Non-Preventable(Retired)**

Majority of turnover is preventable

- Employees are leaving for better opportunities and career growth
- Personal reasons indicate possible work life balance issues

Recommendations

- Improve career growth and promotion paths
- Review salary and benefits competitiveness
- Introduce flexible work and employee support program
- Implement retention strategies

```sql
SELECT
    t.reason_for_leaving,
    COUNT(*) AS "Turnover Frequency"
FROM turnover t
GROUP BY t.reason_for_leaving
ORDER BY "Turnover Frequency" DESC;
```

	reason_for_leaving text	Turnover Frequency bigint
1	Personal	11
2	Found Another Job	7
3	Career Growth	5
4	Retired	5

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Employee Performance Analysis

Goal: Evaluate employee performance across different departments and identify areas where performance can be improved.

How many employees have left the company?

Out of 60 employees, 28 have left, resulting in an attrition rate of 46.67%. This suggests a significantly high turnover level, highlighting the need for immediate retention strategies

```sql
SELECT /*count*/
    count (employee_id) as "Employee left"
FROM turnover;
```

	reason_for_leaving text	Turnover Frequency bigint
1	Personal	11
2	Found Another Job	7
3	Career Growth	5
4	Retired	5

Actionable Recommendations

Conduct Attrition Root Cause Analysis
Analyze exits by:

- Department
- Tenure
- Job role

Use exit interviews to identify key drivers

Focus on High-Turnover Departments
Prioritize intervention in departments with:

- Highest attrition rates

Review:

- Management style
- Work environment
- Team structure

Review Compensation & Benefits

- Benchmark salaries internally and externally
- Address pay gaps and fairness issues

How many employees have a performance score of 5.0

- The company has a **limited pool of top talent**
- Majority of employees fall within **average performance levels**
- Highlights an opportunity to **develop more high performers**

▶ **Recommendation**

- Invest in **training and upskilling programs**
- Implement **performance-based incentives**
- Encourage **knowledge sharing**

```sql
-- How many employees have a performance score of 5.0
SELECT
      performance_score,
      count(distinct employee_id) as " Number of employees"
from performance
where performance_score = 5.0
group by 1

-- -- How many employees have a performance score below 3.5
SELECT
    employee_id,
```

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performance_score numeric (2,1)	Number of employees bigint
5.0	9

How many employees have a performance score below 3.5

75% of employees are performing below a score of 3.5, indicating a widespread performance challenge across the organization. This suggests systemic issues in performance management, employee development, or engagement, requiring immediate intervention through training, improved evaluation systems, and stronger managerial support.

- **1. Upskilling & Training Programs**
  - Identify skill gaps across departments
  - Implement targeted training to improve employee capability

- **2. Strengthen Performance Management**
  - Set **clear KPIs and expectations**
  - Introduce **regular performance reviews (quarterly)**

- **5. 📊 Review Evaluation System**
  - Ensure performance ratings are:
    - Objective
    - Consistent
  - Address possible **rating bias or inflation**

```sql
-- How many employees have a performance score below 3.5
SELECT
    COUNT(distinct employee_id) AS employees_below_3_5
FROM performance
WHERE performance_score < 3.5;
```

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employees_below_3_5 bigint
45

Which department has the most employees with a performance of 5.0 / below 3.5?

- **Trends & Patterns**
- **High concentration of top performers in one department**
  - Performance excellence is **not evenly distributed** across the company
- **Marketing stands out as a performance hub**
  - Likely benefits from:
    - Strong leadership
    - Clear KPIs
    - Effective team collaboration
- **Contrast with other departments**
  - Other departments may lack:
    - Motivation systems
    - Performance clarity
    - Proper support structures
    - **Recommendations**
- **Benchmark Marketing's Success**
- Identify what Marketing is doing right:
  - Management style
  - Incentive structure
  - Workflow processes
- **Maintain Marketing Performance**
- Continue investing in:
  - Talent development
  - Performance incentives
- Ensure top performers are **retained and recognized**
  -

```sql
-- Which department has the most employees with a performance of 5.0 / below 3.5?
select
    department_id,
    count(*) as performance_score
from performance
where performance_score = 5.0
group by department_id
order by performance_score DESC
limit 1
```

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	department_id integer	performance_score bigint
1	3	5

Which department has the most employees with a performance below 3.5?

- **Nearly half of the workforce (47%) are low performers**

Business Risks

Overdependence on a small group of high performers

Potential low morale among struggling employees

Higher turnover risk (low performers may leave or be disengaged)Increased management pressure

- **Recommendations**
- 1. Immediate intervention (Critically)Identify the 28 employees, Place them on Performance Improvement Plans (PIPs)
- 2. Training & development
- Focus on:
- Skill gaps, Role-specific training
- 3. Manager accountability
- Evaluate managers of low-performing teams
- Strengthen supervision and feedback
- 5. Balance workloadEnsure top performers are not overburdened

```sql
---- Which department has the most employees with a performance below 3.5?

select
    department_id,
    count(*) as performance_score
from performance
where performance_score < 3.5
group by department_id
order by performance_score DESC
```

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department_id integer	performance_score bigint
3	28

What is the average performance score by department?

- Department performance is generally strong and consistent (≈4.0+)Dept 3 leads, while Dept 2 lags slightly. Small gap suggests uniform performance, but underlying data shows hidden performance imbalance

- **Recommendations**
- Focus on: Replicating top department success Investigating lower-performing teams

- 2. Benchmark Department 3 Identify what they are doing right example Leadership style, Processes, Employee engagement. Replicate across other departments

- 3. Focus on Department 2 Investigate: Skil, gaps Manager effectiveness, Workload issues.

- 4. Strengthen consistency
- Standardize Training, KPIs, Performance reviews across departments

```sql
-- What is the average performance score by department?
 SELECT
    department_id,
    round(avg(performance_score),2) as " Avg performance"
from performance
group by 1
order by department_id desc
```

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	department_id integer	Avg performance numeric
1	4	4.05
2	3	4.13
3	2	4.00
4	1	4.10

Salary Analysis

What is the total salary expense for the company?

- Total Salary: $4,850,000
- Risk:
- Nearly half of the salary may be going to low performers.
- Opportunity: Optimize training, performance incentives, and workforce distribution
- Goal: Maximize ROI (Return on Investment) on salary expense while retaining top talent

- **Recommendations**

- link compensation to performance
- Introduce performance-based bonuses or incentives
- Reward high performers (5.0)Upskill low performers
- Reduce risk of salary underutilization.
- Invest in training programs
- Review staffing efficiency
- Evaluate team size vs output
- Consider redeployment or role optimization
- Align salary budget with productivity
- Focus salary growth on strategically important roles
- Ensure ROI on salary spend

```sql
-- What is the total salary expense for the company?
select to_char(sum(salary_amount),'$999,999,999') as "Total salary expense"
from salary;
```

Data Output Messages Notifications

Showing rows: 1 to 1 Page No: 1

Total salary expense text
$ 4,850,000

What is the average salary by job title?

- Salaries are relatively high and close togetherBut performance is not equally strong
- This suggests low ROI on salary spend

- **Recommendations**

- Review pay equity
- Ensure employees in similar roles are fairly compensated
- Support lower-level roles
- Consider benefits, bonuses, or incremental raises to improve retention
- Align salary with performance
- High performers should be rewarded to reduce turnover risk
- Benchmark with industry
- Compare salaries externally to stay competitive

```sql
select
    e.job_title,
    round(avg(s.salary_amount),2) as "avg salary"
from salary s
join employee e on e.employee_id = s.employee_id
group by 1
order by "avg salary" desc;
```

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	job_title character varying (30)	avg salary numeric
1	Sales Representative	84285.71
2	HR Specialist	81818.18
3	Engineer	80000.00
4	Sales Manager	80000.00
5	Marketing Specialist	77857.14

How many employees earn above 80,000?

- 26 employees earn above $80K, representing a high-value segment of the workforce. While this reflects strong senior talent presence, it may also indicate salary concentration and potential pay gaps that should be managed for fairness and retention.

- **Business Risks**
- Retention risk at the top
- High earners are valuable and may leave for better offers if not engaged.
- Morale concerns
- Large gaps between high and low earners can affect team motivation.
- Budget pressure
- These 26 employees likely contribute significantly to total salary expenses.

- **Recommendations**
- The organization should focus on retaining top performers, ensuring pay equity, aligning compensation with performance, and building a strong talent pipeline to maintain balance and sustainability.

```sql
-- How many employees earn above 80,000?
select
    count(distinct employee_id) as "salary above 80k"
from salary
where salary_amount > 80000
```

Data Output Messages Notifications

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	salary above 80k bigint
1	26

How does performance correlate with salary across departments?

- **No strong positive relationship** between performance and salary
- Trend shows a **slight negative correlation**
- Higher performance does **not** equal higher pay

▶ **Implications**

- Pay is **role-based, not performance-based**
- Risk of **employee dissatisfaction**
- Potential **retention issues** in high-performing teams

▶ **Recommendations**

- Introduce **performance-based incentives**
- Conduct **salary equity review**
- Reassess **Marketing compensation**
- Align pay with **performance + business impcts**

```sql
SELECT
    e.department_id,
    round(corr(p.performance_score, s.salary_amount):: numeric, 2) as perf_salary_corr
from employee e
join performance p
    on e.employee_id = p.employee_id
join salary s on e.employee_id = s.employee_id
group by e.department_id
order by perf_salary_corr
desc
```

Data Output Messages Notifications

SQL

Showing rows: 1 to 4 Page No: 1 of 1

	department_id integer	perf_salary_corr numeric
1	1	0.21
2	4	0.16
3	2	0.10
4	3	-0.07

In conclusion

- NextGen Corp. shows a clear misalignment between **performance, salary, and employee retention**. High-performing employees are not consistently rewarded, while turnover remains elevated in key areas
- indicating risks to morale, productivity, and long-term growth.

- **Recommendations**
- **Align Pay with Performance**
  Introduce bonuses and salary adjustments tied to performance scores.
- **Address Turnover Hotspots**
  Identify high-exit departments and resolve root causes (low pay, workload, limited growth).
- **Review Salary Structure**
  Ensure fair and competitive compensation across departments, especially for high performers.
- **Strengthen Retention Strategies**
  Invest in career development, recognition programs, and employee engagement.

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- **Bottom Line:**
  To stay competitive, NextGen Corp. must shift from **role-based rewards** to a **performance-driven and employee-centric strategy**.

THANK YOU